EXPERIMENT 18

AIM:

To implement the K-Nearest Neighbours (KNN) classification algorithm to classify biscuits based on their weight and sweetness.

CODE:

```
import numpy as np
import pandas as pd

# Load the dataset
df = pd.read_csv('Fruit_KNN.csv')
df.info()

# Count the number of each fruit
print(df.Fruit.value_counts())

# Display the first five rows
print(df.head())

# Encode the Fruit column
from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()
df['Fruit'] = encoder.fit_transform(df['Fruit'])
```

```
print(df.head())

# Separate features and label
features = df.iloc[:, :-1].values
label = df.iloc[:, 2].values

# Import required libraries
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

# Split the dataset
xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=0.2, random_state=42
)

# Create and train the KNN model
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)

# Evaluate the model
print("Training Accuracy:", model_KNN.score(xtrain, ytrain))
print("Testing Accuracy:", model_KNN.score(xtest, ytest))

# Generate the confusion matrix
from sklearn.metrics import confusion_matrix
```

```
print(confusion_matrix(label, model_KNN.predict(features)))


# Generate the classification report
from sklearn.metrics import classification_report


print(classification_report(
   label,
   model_KNN.predict(features)
))
```

OUTPUT:

1. Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Weight     30 non-null     int64
 1   Sweetness  30 non-null     float64
 2   Fruit      30 non-null     object
dtypes: float64(1), int64(1), object(1)
memory usage: 852.0+ bytes
```

2. Fruit Class Distribution

```
Fruit
Apple     10
Banana    10
Orange    10
Name: count, dtype: int64
```

3. Original Dataset – First Five Records

	Weight	Sweetness	Fruit
0	120	6.0	Apple
1	125	6.2	Apple
2	130	6.4	Apple
3	135	6.5	Apple
4	140	6.7	Apple

4. Label Encoding

	Weight	Sweetness	Fruit
0	120	6.0	0
1	125	6.2	0
2	130	6.4	0
3	135	6.5	0
4	140	6.7	0

5. KNN Model

```
KNeighborsClassifier()
```

6. Training and Testing Accuracy

```
Training Accuracy: 0.9166666666666666
Testing Accuracy: 0.6666666666666666
```

7. Confusion Matrix

```
array([[ 9,  0,  1],
       [ 3,  7,  0],
       [ 0,  0, 10]])
```

8. Classification Report

	precision	recall	f1-score	support
0	0.75	0.90	0.82	10
1	1.00	0.70	0.82	10
2	0.91	1.00	0.95	10
accuracy			0.87	30
macro avg	0.89	0.87	0.86	30
weighted avg	0.89	0.87	0.86	30

RESULT:

The KNN classification algorithm was successfully implemented, and biscuit classification performance was evaluated using accuracy, a confusion matrix, and a classification report.